

Label Quality

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Inspect Before Modeling

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Classification or Regression?

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Interpret agreement for Label definitions and disagreement. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Label definitions and disagreement. Proposed explanation: “Agreement is accuracy”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply the process independently

New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear. P has an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has an object inside. Count agreement including review. Apply v1 to all five. Then report agreement on the original pairs excluding any review status, with its denominator and limitation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute agreement and explain why it does not establish correct labels.

6. Solve the given case

Two annotators disagree on three of ten leaves, especially mild spotting.

Improve label quality.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Two labelers assign different species to the same image. Should you silently pick the label that helps the model score? Describe a better process.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 6 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute agreement and explain why it does not establish correct labels.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Use the boundary and review rules

Under rule v1, label a wheel touching the boundary, an object entirely outside a fully visible lane, and a partly hidden lane. Explain the evidence for each.

My reasoning and evidence

13. W2 Interpret agreement

Explain the output 2 4 for A through D. State the fraction and percent, whether review is included, and why this is not model accuracy.

My reasoning and evidence

14. W3 Write a decision log

For B and C, record original labels, visible evidence, rule version and reviewed decision. Explain why the original labels should remain available.

My reasoning and evidence

15. W4 Apply the process independently

New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear. P has

an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has an object inside. Count agreement including review. Apply v1 to all five. Then report agreement on the original pairs excluding any review status, with its denominator and limitation.

My reasoning and evidence

16. W5 Draft and test a label rule

Write a short v1 labeling instruction covering blocked, clear, the touching boundary and review. Add one boundary example. Explain why three people voting clear cannot settle a hidden lane.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Label Quality

Write a label rule that resolves the visible boundary case.

Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review. X has one visible spot; Y has none; Z is half hidden.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Label Quality

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Apply visible rule

X is spotted; Y is healthy under this exercise's definition.



2. Handle uncertainty

Z requires review; hidden tissue cannot be treated as visibly spot-free.



3. Review disagreement

If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

CHANGE ONE THING

Z receives a new fully visible image with no brown spots.

Explain the changed result using the matching lesson.

Why should two annotators not vote away Z's missing evidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Write a label rule that resolves the visible boundary case.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

1. Apply visible rule
2. Review disagreement
3. Handle uncertainty

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Review disagreement If an annotator labels X healthy, compare the image and declared spot rule before changing the label. Read the picture in order. If an annotator labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Z receives a new fully visible image with no brown spots. Which conclusion is supported by the supplied facts?

1. Label the new observation healthy under the toy rule.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Label the new observation healthy under the toy rule. The missing visual evidence is now supplied; the earlier uncertain image was not proof of health.

Repair a missing building block

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
